How-To Library

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Converting Relational Input into Hierarchical Output using Microsoft Azure Cosmos DB SQL API Connector

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Abstract

When you want to read data from a relational source and write data to Microsoft Azure Cosmos DB SQL API, you can use the Hierarchical Builder transformation to convert relational input into hierarchical output. The transformation processes a relational input from the upstream transformation and provides JSON output to the downstream transformation. This article describes how to use the Hierarchical Builder transformation with Microsoft Azure Cosmos DB SQL API Connector, to convert relational input into hierarchical output.

Supported Versions

- Cloud Data Integration

Table of Contents

Overview. 2
Before You Begin. 3
Create a Hierarchical Schema. 3
Add a Source Transformation. 4
Add a Joiner Transformation. 4
Add a Hierarchical Builder Transformation to the Mapping. 5
Associate the Hierarchical Schema with the Hierarchical Builder Transformation. 5
Configure the Field Mapping to Select Schema Elements. 6
Add an Expression Transformation. 6
Add a Target Transformation. 7
Configure a Mapping Task. 8

Overview

When you want to write data from relational sources to Microsoft Azure Cosmos DB SQL API, you can use the Hierarchical Builder transformation to convert relational input into hierarchical output.

Use a schema file to create a hierarchical schema that defines the expected hierarchy of the output data. Create and configure the Hierarchical Parser transformation in Data Integration and add it to a mapping. Then, associate the schema definition file with the Hierarchical Builder transformation. When you run the mapping, the Hierarchical Parser transformation converts the relational input into the hierarchical output based on the schema file.

To configure a Hierarchical Parser transformation in a Microsoft Azure Cosmos DB SQL API mapping, perform the following steps:

1. Configure a JSON schema definition file on the Secure Agent.
2. Add a Source transformation.
3. Add a Joiner transformation (Optional).
4. Add a Hierarchical Builder transformation to the mapping.
   - Associate the JSON schema definition file with the Hierarchical Builder transformation.
   - Configure the field mapping and link the relational fields to schema elements that provide the hierarchical output.
5. Add an Expression transformation.

6. Add a Target transformation.
7. Run the mapping or the mapping task.

Before You Begin

Verify that you have the permission to use the following licenses and packages before you create a mapping:

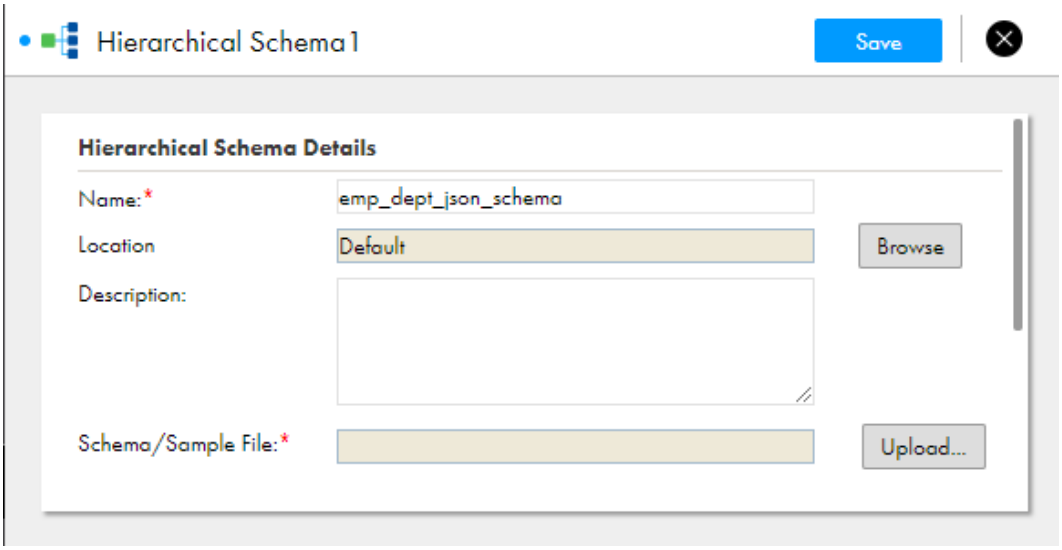
- DT XML Processing (License)
- Data Transformation
- UTD for Hierarchy

Create a Hierarchical Schema

You must associate a JSON schema definition file with the Hierarchical Builder transformation to define the schema of the Microsoft Azure Cosmos DB SQL API target object. The Hierarchy Builder transformation converts hierarchical input based on the JSON schema definition file.

1. In Data Integration, click **New** > **Components** > **Hierarchical Schema**.

   The **Hierarchical Schema 1** page appears.



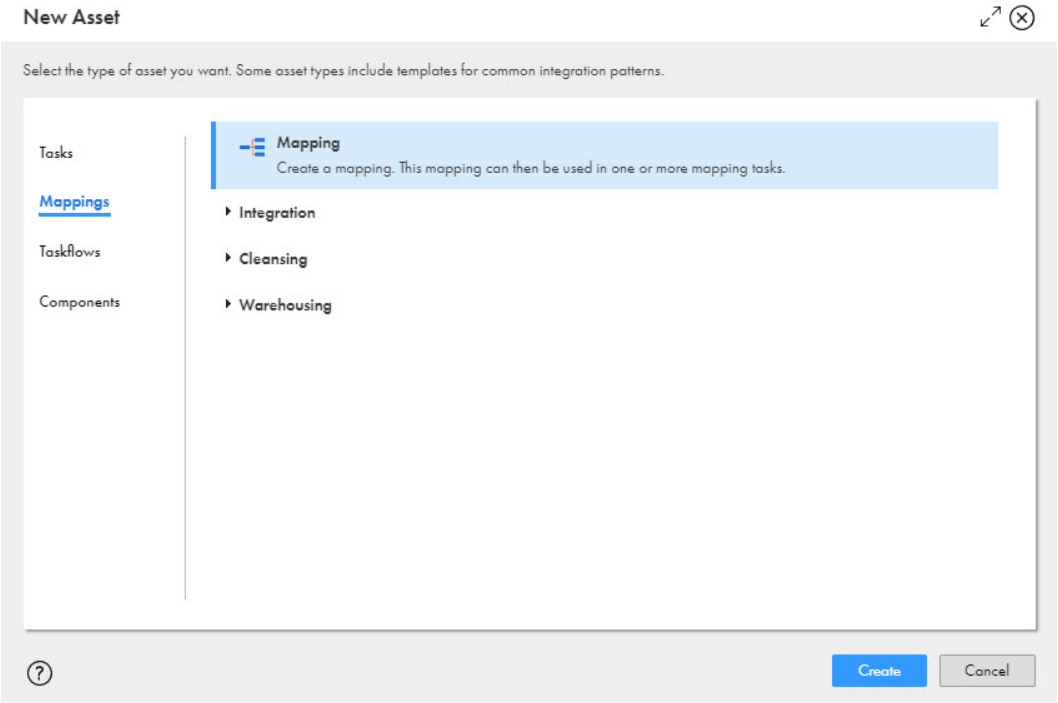
2. Enter a name for the hierarchical schema and optionally enter a description.
3. To select a schema, click **Upload**, browse for a schema file, and click **OK**.
4. To save the hierarchical schema, click **Save**.

Add a Source Transformation

Add one or more relational source objects and configure the source properties.

1. To create a mapping, click **New** > **Mapping**.

   The following image shows the **New Asset** dialog box with Mapping selected:



2. Click **Create**. The **Mapping Designer** appears with a new mapping displayed in the mapping canvas.
3. On the mapping canvas, click the Source transformation.
4. Specify the name and description of the source object.
5. Click **Source** and select an active Server connection.
6. Select `Single Object` as Source Type.
7. Configure advance properties for the source objects as required.
8. Select a Cosmos DB collection as the object from where you want to read the data.
9. Repeat the steps to add more relational sources to the mapping, if required.

Add a Joiner Transformation

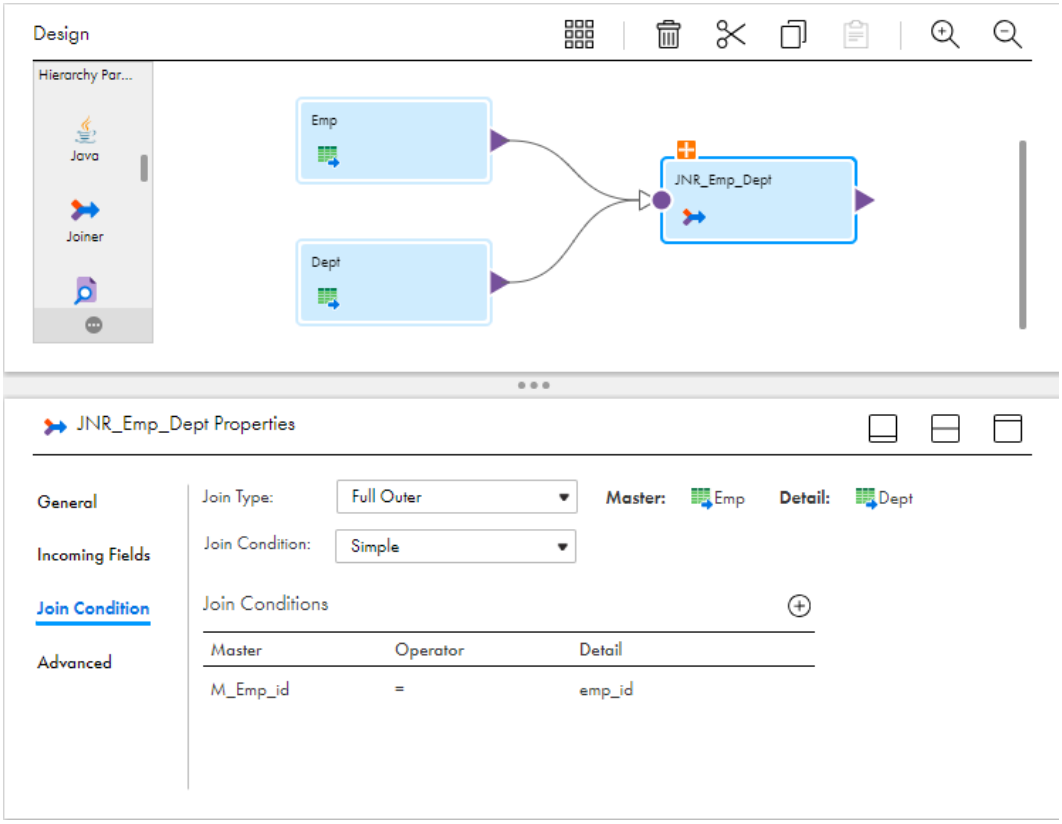
Use a Joiner transformation to join data from two related heterogenous sources.

1. In the **Transformation palette**, drag a Joiner transformation onto the mapping canvas.
2. Connect an upstream transformation that represents one data set to the Master group of the Joiner transformation.

   To improve job performance, use the transformation that represents the smaller data set.
3. Connect an upstream transformation that represents the other data set to the Detail group of the Joiner transformation.
4. On the **General** tab, enter a name and optional description for the transformation.

5. On the **Incoming Fields** tab, configure the field rules that define the data that enters the transformation.
6. On the **Field Mappings** tab, select the **Join Type**.
7. To create a join condition, the **Join Condition**, select Simple to create join conditions.

   The following image shows the join condition between the `Emp` and the `Dept` tables:



Add a Hierarchical Builder Transformation to the Mapping

You must add a Hierarchical Builder transformation to the mapping to convert the relational input into hierarchical output. The transformation processes the relational input from the upstream transformation and provides JSON output to the downstream transformation. The Hierarchy Builder transformation produces hierarchical output based on the JSON schema definition file that you associate with the transformation.

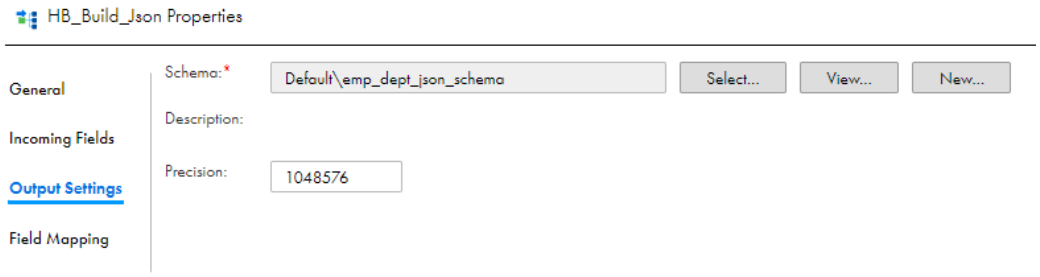
1. Drag the Hierarchical Builder transformation from the **Transformation** palette into the editor. Click the **Properties** panel to configure the properties.
2. Establish a connection between the Joiner and the Hierarchical Builder transformation.

Associate the Hierarchical Schema with the Hierarchical Builder Transformation

Select the sample schema on the **Output Settings** tab of the **Properties** panel.

1. Select the Hierarchical Builder transformation and click the **Output Settings** tab under the **Properties** panel.
2. Select the schema you created to convert the relational input into the hierarchical output.

The following image shows the **Output Settings** tab:



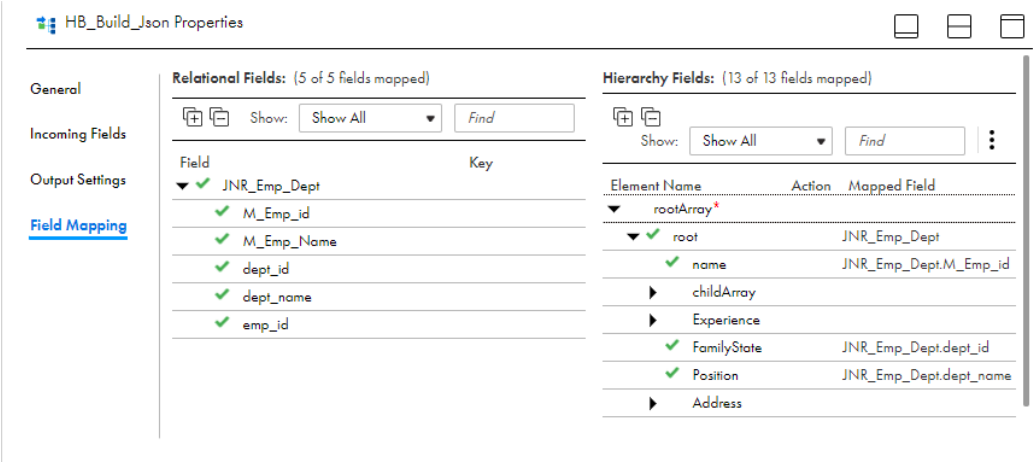
3. To save the schema and include it in the Hierarchical Builder transformation, click **Save**.

Configure the Field Mapping to Select Schema Elements

Configure the field mapping in the Hierarchy Builder transformation to define which relational elements link to schema elements to provide the hierarchical output. You use the field mapping editor to link relational fields to schema elements. The field mapping editor displays the transformation input elements and output fields. To the left, the editor shows the schema elements. To the right, the editor shows the relational output fields.

- In the **Properties** panel of the Hierarchical Builder transformation, click the **Field Mapping** tab.

  The following image shows the field mapping editor:



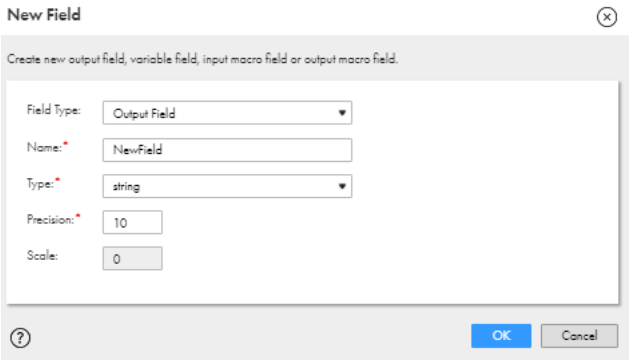
Add an Expression Transformation

You must place an Expression transformation after the Hierarchical Builder transformation to write data to Microsoft Azure Cosmos DB SQL API target.

1. Drag the Expression transformation from the **Transformation** palette into the editor.
2. On the **General** tab, enter a name and description.
3. Draw a link to connect the Hierarchical Builder transformation.
4. On the **Incoming Fields** tab, you can configure the field rules that define the data that enters the transformation.
5. On the **Expression** tab, create a new field. Select the **Field Type** as Output Field and **Type** as string.

   The **New Field** dialog box appears.

The following image shows the **New Field** dialog box:

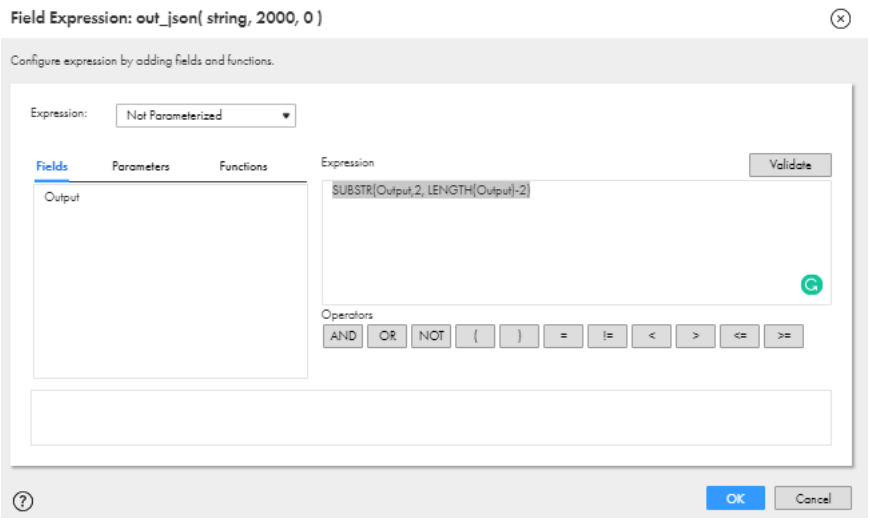


6. Click **OK**.
7. Click **Configure** to configure the field expression.

   The **Field Expression** dialog box appears.
8. Select the **Expression** as Not Parameterized.
9. Add the following expression in the **Expression** field to remove the additional square braces, "[" "]" in the relational target data:

   ```
   SUBSTR(Output,2, LENGTH(Output)-2)
   ```

   The following image shows the **Field Expression** dialog box:



10. Click **Validate** to validate the expression.
11. Click **OK**.
12. Draw a link to connect the Expression transformation to the target.

Add a Target Transformation

Use the Target transformation to define the Microsoft Azure Cosmos DB SQL API target in the mapping.

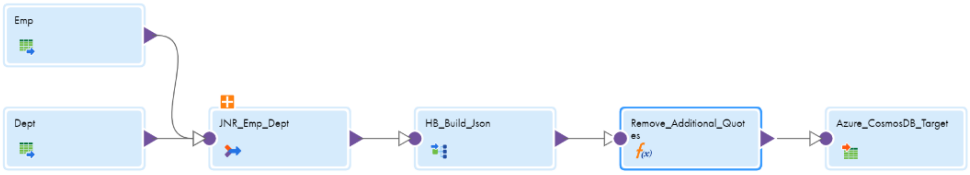
1. Drag a Target transformation onto the mapping canvas.
2. Click the **Target** tab under the **Properties** panel.

3. Select an active Microsoft Azure Cosmos DB SQL API connection and select a Microsoft Azure Cosmos DB SQL API target object.
4. Select `None` as **Format Type** in **Formatting Options**.
5. Establish a connection between the Expression transformation and the target.
6. Click **Save**.

Configure a Mapping Task

You can save and run the mapping directly or configure a mapping task, add the configured mapping, and run the mapping.

1. To create a mapping task, click **New** > **Tasks**, and then click **Create**.
2. In the **New MappingTask** window, provide a task name, and provide the configuration details.
3. Select the configured mapping, and click **OK**.
1. The following image shows the configured mapping:



4. Configure a schedule and advanced options, if required.
5. Click **Save**.

   You can run the mapping task manually or on a schedule.

Author

Ripa Bhagawati

Acknowledgements

The author would like to acknowledge Sreehari Desai, for his technical assistance.